Consistent Hashing

Consistent hashing is a technique for distributing keys across a set of servers so that adding or removing a server moves only a small fraction of the keys. In a simple hash scheme, server = hash(key) % N. If N changes, almost every key maps to a different server, which causes a large amount of data movement and cache misses.

The Hash Ring

In consistent hashing, both servers and keys are hashed onto a circular space, often called the hash ring. The output range of the hash function wraps around, so the largest value is followed by the smallest. To find the server for a key, start at the key's position on the ring and walk clockwise until you reach the first server. That server owns the key.

Virtual Nodes

Placing each physical server at only one point on the ring can produce uneven load. Virtual nodes solve this by giving each server many points on the ring, for example 100 or 200 replicas. The keys are then spread more evenly, and when a server is removed, its load is distributed across many other servers instead of a single neighbor.

Adding and Removing Servers

When a server is added, it takes over only the keys between its position and the previous server on the ring. All other keys stay where they are. When a server is removed, its keys move to the next server clockwise. On average, only about K/N of K keys need to move when a new server joins a cluster of N servers.

Where It Is Used

Consistent hashing is used in distributed caches such as Memcached client libraries, in distributed databases like Amazon DynamoDB and Apache Cassandra, and in load balancers. It is a common topic in system design interviews because it shows how to scale a cluster without a full reshuffle of data.